Implementing Gradient Descent

From slope to the next point **Worked answers**

Name _____ Date _____

01 One Update

Shared Data and Model

Input x	-1	1
Target y	-2	2

$f_w(x) = wx$ $m = 2$ $b = 0$

Fix $b = 0$ and change only w One cross-section of the 3D cost surface

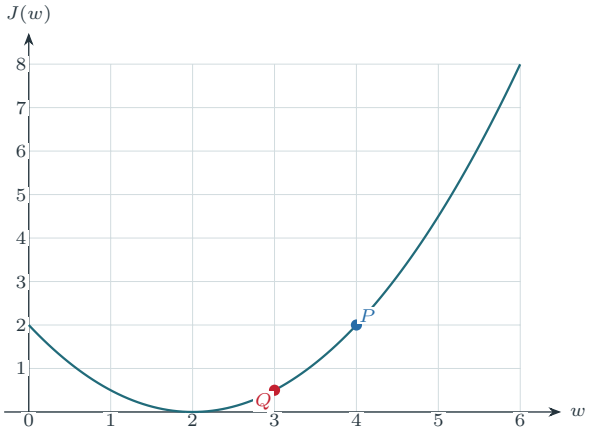
1 Calculate the predictions and cost at $w = 4$

$$J(w) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_w(x^{(i)}) - y^{(i)}\right)^2 = \frac{(-w+2)^2 + (w-2)^2}{4} = \frac{1}{2}(w-2)^2$$

$f_4(-1) = 4(-1) = -4$, $f_4(1) = 4 \times 1 = 4$

$J(4) = \dfrac{(-4+2)^2 + (4-2)^2}{4} = \dfrac{4+4}{4} = 2$

Blue point P is $(w, J(w)) = (4, 2)$



2 Use the sign of the slope to choose the direction of the update

$$J'(w) = \frac{d}{dw}J(w) = w - 2 \quad (\alpha > 0)$$

Current w	Slope $w-2$	Direction of the next w
0	$0-2=-2$	☐ Decrease ☐ Increase ☐ Unchanged
2	$2-2=0$	☐ Decrease ☐ Increase ☐ Unchanged
4	$4-2=2$	☐ Decrease ☐ Increase ☐ Unchanged

3 Starting at $w = 4$ with $\alpha = 0.5$, make one update and mark the new point Q above

$w_{k+1} = w_k - \alpha J'(w_k)$ current value ⟶ next value

* w_0 is the starting value; w_1 is the first updated value Use the given derivative

$w_{\text{new}} = 4 - 0.5 \times 2 = 3$ $J(w_{\text{new}}) = 0.5$

Question 2 · At $w = 0$, $w_{\text{new}} = 0 - \alpha(-2) = 2\alpha > 0$, so w increases
At $w = 2$, $2 - \alpha \times 0 = 2$ is unchanged; at $w = 4$, $4 - 2\alpha < 4$, so w decreases
Question 3 · $J'(4) = 4 - 2 = 2$ $w_{\text{new}} = 4 - 0.5 \times 2 = 4 - 1 = 3$
$J(3) = \frac{1}{2}(3-2)^2 = \frac{1}{2} \times 1 = 0.5$ Therefore $Q = (3, 0.5)$

Reference · Supplied Gradient Descent algorithm and Intuition slides · Update rule and slope signs through C1 W1 slide 65

02 Learning Rate A

$J(w) = \frac{1}{2}(w-2)^2$ $J'(w) = w - 2$ $w_0 = 6$ $\boxed{\alpha = 3}$

4 Use the same rule to make **five updates**

$$w_{k+1} = w_k - 3(w_k - 2)$$

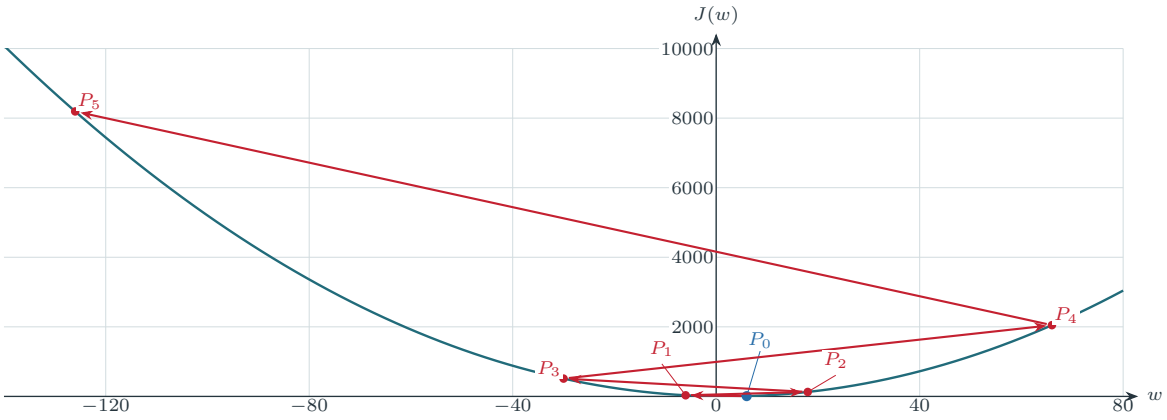
* Use the last value in a row as the current w in the next row

k	Current w_k	Slope $w_k - 2$	Subtract $\alpha(w_k - 2)$	Next w_{k+1}
0	6	$6 - 2 = 4$	$3 \times 4 = 12$	$6 - 12 = -6$
1	-6	$-6 - 2 = -8$	$3 \times (-8) = -24$	$-6 - (-24)$ $= -6 + 24 = 18$
2	18	$18 - 2 = 16$	$3 \times 16 = 48$	$18 - 48 = -30$
3	-30	$-30 - 2 = -32$	$3 \times (-32) = -96$	$-30 - (-96)$ $= -30 + 96 = 66$
4	66	$66 - 2 = 64$	$3 \times 64 = 192$	$66 - 192 = -126$

5 Calculate each cost, plot $P_k = (w_k, J(w_k))$, and draw arrows in order $P_0 \to P_1 \to \cdots \to P_5$

* $J(w_k) = \frac{1}{2}(w_k - 2)^2$ $P_0 = (6, 8)$

$J(w_1)$	$J(w_2)$	$J(w_3)$	$J(w_4)$	$J(w_5)$
$\frac{1}{2}(-6-2)^2$ $= 32$	$\frac{1}{2}(18-2)^2$ $= 128$	$\frac{1}{2}(-30-2)^2$ $= 512$	$\frac{1}{2}(66-2)^2$ $= 2048$	$\frac{1}{2}(-126-2)^2$ $= 8192$



6 Check the **position and cost** after five updates

$w_5 =$ -126 $|w_5 - 2| =$ 128

$J(w_5) = \frac{1}{2}(w_5 - 2)^2 =$ 8192 $J(w_0) = 8$

☐ Closer ☐ Farther away ☐ Almost no movement

Explain why the cost increased even though you moved opposite to the slope

$|w_5 - 2| = |-126 - 2| = |-128| = 128$ $J(w_5) = \frac{1}{2}(-128)^2 = \frac{1}{2} \times 16384 = 8192 > 8$
The first update $w_1 = 6 - 3 \times 4 = -6$ overshoots the minimum $w = 2$ $J(-6) = 32 > J(6) = 8$
Distance to the minimum: $4 \to 8 \to 16 \to 32 \to 64 \to 128$ The updates move farther away on alternating sides: divergence

03 Learning Rate B

$J(w) = \frac{1}{2}(w-2)^2$ $J'(w) = w - 2$ $w_0 = 6$ $\boxed{\alpha = 0.01}$

7 Use the same rule to make **five updates**

$$w_{k+1} = w_k - 0.01(w_k - 2)$$

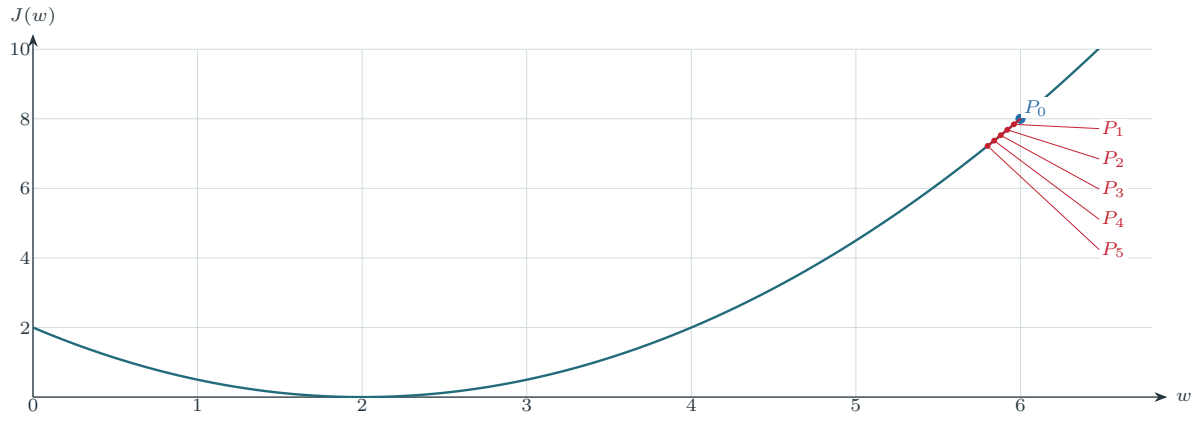
* For B, round each new w to two decimal places before the next row Table and plot values are approximate

k	Current w_k	Slope $w_k - 2$	Subtract $\alpha(w_k - 2)$	Next w_{k+1}
0	6	$6 - 2 = 4$	$0.01 \times 4 = 0.04$	$6 - 0.04 = 5.96$
1	5.96	$5.96 - 2 = 3.96$	$0.01 \times 3.96 = 0.0396$	$5.96 - 0.0396 = 5.9204 \approx 5.92$
2	5.92	$5.92 - 2 = 3.92$	$0.01 \times 3.92 = 0.0392$	$5.92 - 0.0392 = 5.8808 \approx 5.88$
3	5.88	$5.88 - 2 = 3.88$	$0.01 \times 3.88 = 0.0388$	$5.88 - 0.0388 = 5.8412 \approx 5.84$
4	5.84	$5.84 - 2 = 3.84$	$0.01 \times 3.84 = 0.0384$	$5.84 - 0.0384 = 5.8016 \approx 5.8$

8 Calculate each cost, plot $P_k = (w_k, J(w_k))$, and draw arrows in order $P_0 \to P_1 \to \cdots \to P_5$

* $J(w_k) = \frac{1}{2}(w_k - 2)^2$ $P_0 = (6, 8)$

$J(w_1)$	$J(w_2)$	$J(w_3)$	$J(w_4)$	$J(w_5)$
$\frac{1}{2}(5.96-2)^2 = 7.8408$	$\frac{1}{2}(5.92-2)^2 = 7.6832$	$\frac{1}{2}(5.88-2)^2 = 7.5272$	$\frac{1}{2}(5.84-2)^2 = 7.3728$	$\frac{1}{2}(5.8-2)^2 = 7.22$



9 Check the **position and cost** after five updates

$w_5 \approx$ 5.80 $|w_5 - 2| \approx$ 3.80

$J(w_5) = \frac{1}{2}(w_5 - 2)^2 \approx$ 7.22 $J(w_0) = 8$

☐ At the minimum ☐ Stuck in another valley ☐ Still moving slowly

$|w_5 - 2| \approx |5.80 - 2| = 3.80$ $J(w_5) \approx \frac{1}{2}(3.80)^2 = \frac{1}{2} \times 14.44 = 7.22$
The next step is also about $0.01 \times 3.80 = 0.038$ The slope is about 3.80, but small α makes movement slow
The minimum has not been reached Continuing the unrounded update rule approaches $w = 2$ slowly

04 Learning Rate C

$J(w) = \frac{1}{2}(w-2)^2$ $J'(w) = w - 2$ $w_0 = 6$ $\boxed{\alpha = 0.5}$

10 Use the same rule to make **five updates**

$$w_{k+1} = w_k - 0.5(w_k - 2)$$

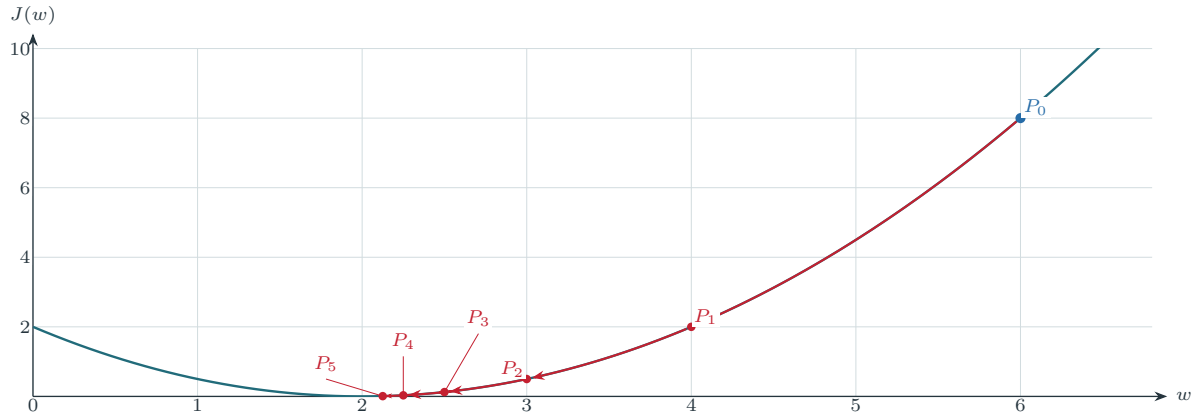
* Use the last value in a row as the current w in the next row

k	Current w_k	Slope $w_k - 2$	Subtract $\alpha(w_k - 2)$	Next w_{k+1}
0	6	$6 - 2 = 4$	$0.5 \times 4 = 2$	$6 - 2 = 4$
1	4	$4 - 2 = 2$	$0.5 \times 2 = 1$	$4 - 1 = 3$
2	3	$3 - 2 = 1$	$0.5 \times 1 = 0.5$	$3 - 0.5 = 2.5$
3	2.5	$2.5 - 2 = 0.5$	$0.5 \times 0.5 = 0.25$	$2.5 - 0.25 = 2.25$
4	2.25	$2.25 - 2 = 0.25$	$0.5 \times 0.25 = 0.125$	$2.25 - 0.125 = 2.125$

11 Calculate each cost, plot $P_k = (w_k, J(w_k))$, and draw arrows in order $P_0 \to P_1 \to \cdots \to P_5$

* $J(w_k) = \frac{1}{2}(w_k - 2)^2$ $P_0 = (6, 8)$

$J(w_1)$	$J(w_2)$	$J(w_3)$	$J(w_4)$	$J(w_5)$
$\frac{1}{2}(4-2)^2 = 2$	$\frac{1}{2}(3-2)^2 = 0.5$	$\frac{1}{2}(2.5-2)^2 = 0.125$	$\frac{1}{2}(2.25-2)^2 = 0.03125$	$\frac{1}{2}(2.125-2)^2 = 0.0078125$



12 Check the **position and cost** after five updates

$w_5 =$ 2.125 $|w_5 - 2| =$ 0.125 $J(w_5) = \frac{1}{2}(w_5 - 2)^2 =$ $\frac{1}{128}$ $J(w_0) = 8$

☐ Closer ☐ Farther away ☐ Almost no movement

Explain why the steps get smaller even though the learning rate is fixed

$|w_5 - 2| = |2.125 - 2| = 0.125 = \frac{1}{8}$
$J(w_5) = \frac{1}{2}(\frac{1}{8})^2 = \frac{1}{2} \times \frac{1}{64} = \frac{1}{128} < 8$, so the point is closer to the minimum
Step sizes: $2 \to 1 \to 0.5 \to 0.25 \to 0.125$ Even with fixed α, a smaller slope gives a smaller step

13 Which case is closest to the minimum after five updates? C

Does a small step alone establish convergence? No

Distance to the minimum: C 0.125 < B about 3.80 < A 128 C is closest